



RippleWealth: AI-Driven Investment Risk Intelligence Platform

Sprint 2 Deliverable:

Computer Science Capstone Project CS 691

Prof. Kshitij Sharma

Spring 2026

Group 8:

Sharvani Kadarla

Deepali Sawant

Jai Sheel Bhasker Paladugu

Ashish Reddy Majjigapu

Agenda

- ❑ Team Member Roles & Responsibilities
- ❑ Improvements from Professor Feedback
- ❑ Project Overview
- ❑ Team Working Agreement
- ❑ Personas
- ❑ Minimum Viable Product (MVP)
- ❑ System Design & Technologies
- ❑ Algorithms Used
- ❑ System Diagrams

- ❑ Sprint 1 Recap
- ❑ Product Backlog
- ❑ Sprint 2 Backlog
- ❑ Test Cases
- ❑ Sprint Metrics
- ❑ Sprint Retrospective
- ❑ Sprint 3 Planning
- ❑ Project Demonstration
- ❑ GitHub Repository & Live Application

Team Member Roles and Responsibilities

Our project success hinges on a clearly defined structure, where each team member's expertise drives specific outcomes.

Sharvani Kadarla
**Project Manager &
Software Developer**

Define project vision, roadmap, and milestones

Plan sprints and coordinate team tasks

Design system architecture and data flow

Ensure integration across frontend, backend, and AI modules

Monitor project progress and resolve blockers

Deepali Sawant
**Frontend & Visualization
Engineer**

Develop interactive UI and dashboards

Build portfolio and knowledge graph visualizations

Ensure clean, user-friendly UI/UX design

Integrate frontend with backend APIs and AI outputs

Jai Sheel Bhasker Paladugu
**AI / Machine Learning
Engineer**

Develop AI models for event impact analysis

Build causal graphs for industry and company effects

Implement LLM-based explainable insights

Run scenario simulations for portfolio risk analysis

Ashish Reddy Majjigapu
**Backend & Data
Engineering Lead**

Develop RESTful APIs using FastAPI

Build data ingestion and preprocessing pipelines

Manage data storage and validation

Support portfolio risk calculations and backend performance

Improvements made from Professor's Feedback

After reviewing the professor's feedback on our sprint evaluation, we identified the need to improve planning, prioritization, and timely task completion.

To address this, we improved our sprint planning process by clearly defining user stories, assigning responsibilities, and prioritizing tasks earlier in the sprint. We also introduced better progress tracking and internal deadlines to avoid delays in deliverables.

Additionally, the team strengthened communication and coordination through regular meetings and status updates, ensuring that tasks are completed on time and that any blockers are resolved quickly.

These improvements helped the team maintain better alignment, accountability, and more efficient sprint execution for the RippleWealth project.

Problem Statement

- ❑ Modern investment platforms are reactive — they alert investors only after stock prices change
- ❑ They do not consider early real-world signals like supply chain disruptions, weather events, labor strikes, or logistics delays
- ❑ Current tools focus mainly on financial indicators and ignore non-financial risk triggers
- ❑ Investors lack clear explanations of why their portfolio risk is increasing
- ❑ As a result, users cannot anticipate risk or make proactive portfolio decisions

Problem Description

RippleWealth is an AI-driven investment risk intelligence platform that anticipates market risks by detecting real-world disruptions—such as weather events, supply chain delays, and labor strikes—before they appear in stock prices. Unlike traditional investment tools that react only after price changes, RippleWealth provides early insights to help investors understand potential portfolio impacts.

Using a multi-agent AI architecture and LLM-based causal reasoning, the platform maps how these events propagate across industries and impact companies and portfolios. Through an interactive Knowledge Graph and explainable risk metrics like Value-at-Risk (VaR), RippleWealth enables proactive, transparent, and data-driven portfolio risk management.

Team Working Agreement

Participation

- ❑ Attend scheduled meetings on time
- ❑ Come prepared with progress updates
- ❑ Inform the team leader in advance if absent

Meetings

- ❑ Weekly meetings via Zoom / Google Meet
- ❑ Team leader schedules and shares meeting links
- ❑ Meeting minutes recorded after each session

Work Quality & Collaboration

- ❑ Share work early for feedback
- ❑ Maintain professionalism and accountability
- ❑ Ensure alignment with project goals

Work Division & Ownership

- ❑ Tasks assigned during sprint planning
- ❑ Equal responsibility among team members
- ❑ Meet agreed deadlines and communicate blockers early

Communication Tools

- ❑ Zoom / Google Meet – Sprint meetings
- ❑ WhatsApp – Quick updates and urgent communication
- ❑ Google Docs & Slides – Collaborative deliverables

Team Transparency

- ❑ Cameras encouraged during meetings
- ❑ Maintain open communication and team trust

User Persona 1: Ethan Miller – Retail Investor (Young Professional)

Ethan represents our younger target user: a data-driven professional eager for sophisticated, simple tools to manage his growing portfolio.



DEMOGRAPHICS

Name: Ethan Miller
Age: 29
Location: Austin, Texas
Job: Software Engineer
Salary: \$95,000–\$120,000/year
Family: Lives with partner
Investment Level: Beginner to Intermediate Investor

GOALS

- ☐ Understand Value-at-Risk clearly
- ☐ Simulate market crash scenarios
- ☐ Get simple portfolio risk scores
- ☐ Improve long-term returns
- ☐ Make safer investment decisions

FRUSTRATIONS

- ☐ Cannot clearly measure portfolio risk
- ☐ Hard to understand downside scenarios
- ☐ Too many scattered finance tools
- ☐ Confusing probability models
- ☐ Lacks simple visual risk insights

INTERESTS

- ☐ Personal finance and investing
- ☐ Portfolio tracking apps
- ☐ Market trend analysis
- ☐ Tech and automation tools
- ☐ Financial podcasts
- ☐ Long-term wealth building

User Persona 2: Sophia Chen – Financial Risk Analyst (Professional User)

Sophia represents our advanced professional user: an expert operating in a high-stakes, data-intensive environment where deep analytics and systemic accuracy are critical.



DEMOGRAPHICS

Name: Sophia Chen
Age: 38
Location: New York, NY
Job: Financial Risk Analyst
Salary: \$130,000–\$160,000/year
Family: Married, one child
Investment Level: Advanced / Professional

GOALS

- ☐ Run large-scale Monte Carlo simulations
- ☐ Identify systemic risk pathways
- ☐ Model inter-company relationships
- ☐ Generate client risk reports quickly
- ☐ Improve predictive accuracy

FRUSTRATIONS

- ☐ Manual scenario modeling is time-consuming
- ☐ Tools lack integrated causal analysis
- ☐ Hard to map indirect market impacts
- ☐ Needs faster simulation turnaround
- ☐ Limited explainability in some models

INTERESTS

- ☐ Risk modeling techniques
- ☐ Market stress testing
- ☐ Portfolio optimization
- ☐ Quantitative finance
- ☐ Economic indicators
- ☐ Financial data visualization

User Persona 3: Carlos Ramirez – Business Owner / Wealth Planner

Carlos represents our entrepreneur user: a strategic investor who manages complex business and personal holdings and needs to see the "cause-and-effect" of global economic events on his investments.



DEMOGRAPHICS

Name: Carlos Ramirez

Age: 45

Location: Atlanta, Georgia

Job: Small Business Owner

Salary: \$110,000–\$150,000/year

Family: Married

Investment Level: Moderate Investor

GOALS

- ☐ Understand event → market → portfolio impact
- ☐ Visualize causal financial relationships
- ☐ Stress-test business investments
- ☐ Reduce exposure to high-risk sectors
- ☐ Make resilient investment plans

FRUSTRATIONS

- ☐ Unsure how world events affect investments
- ☐ No clear impact chain visibility
- ☐ Market shocks feel unpredictable
- ☐ Hard to evaluate sector dependencies
- ☐ Lacks scenario-based planning tools

INTERESTS

- ☐ Business growth strategy
- ☐ Market and sector trends
- ☐ Supply chain news
- ☐ Financial planning tools
- ☐ Diversified investments
- ☐ Economic forecasts

Minimal Viable Product (MVP)



- **Alternative Data Engine:** monitors news, weather, supply-chain, and labor disruptions using keyword and NLP filters with public data APIs.
- **Causal Chain Prototype:** links Event → Sector → Company → Portfolio so users can trace why risk changes.
- **Portfolio Risk Dashboard:** shows Value-at-Risk and portfolio risk scores using basic Monte Carlo simulation.
- **Explainable Insights:** generates plain-English narratives that connect disruptions to portfolio exposure.

Core Technologies Powering Our Platform

AI / Analytics

- Python, Monte Carlo simulation, Value-at-Risk, and keyword NLP for early-event detection and risk modeling.

Backend APIs

- FastAPI powers REST endpoints for portfolio data, risk scoring, and frontend integration.

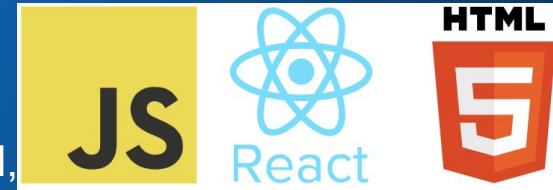
Storage Layer

- SQLite plus CSV and mock API ingestion support user portfolios, event storage, and preprocessing.

Frontend

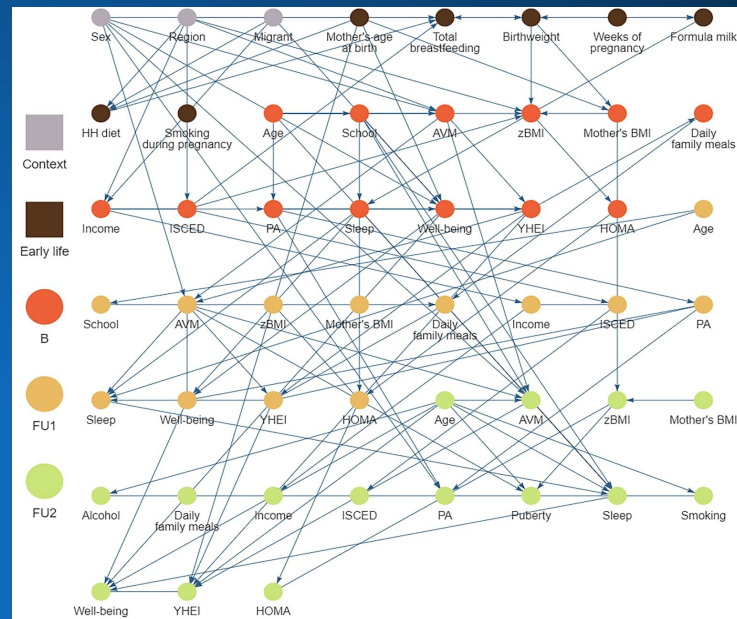
- React.js, JavaScript, HTML5, and CSS3 support the dashboard, interaction flow, and visual components.

Graph technology: Neo4j supports impact tracing and causal relationship visualization across sectors and companies



Algorithms

- 1. Event Detection:** the system monitors alternative data sources such as news, weather, supply-chain updates, and labor signals to identify early butterfly events through keyword and NLP filtering.
- 2. Causal Mapping:** once an event is detected, a causal graph links the disruption to affected sectors, companies, and user portfolios so the platform can trace direct and indirect exposure.
- 3. Risk Simulation:** Monte Carlo simulation runs multiple portfolio scenarios to estimate Value-at-Risk and measure how uncertainty changes after the event enters the system.
- 4. Explainable Output:** the reasoning layer converts model output into plain-English insights that tell the investor what happened, where the impact flows, and why portfolio risk increased.



Event Inputs

News feeds, weather alerts, strikes, and logistics delays provide the early non-financial signals used by the platform.

Graph Logic

Neo4j-style relationship tracing maps how one disruption spreads from event to sector to company and then to a holding.

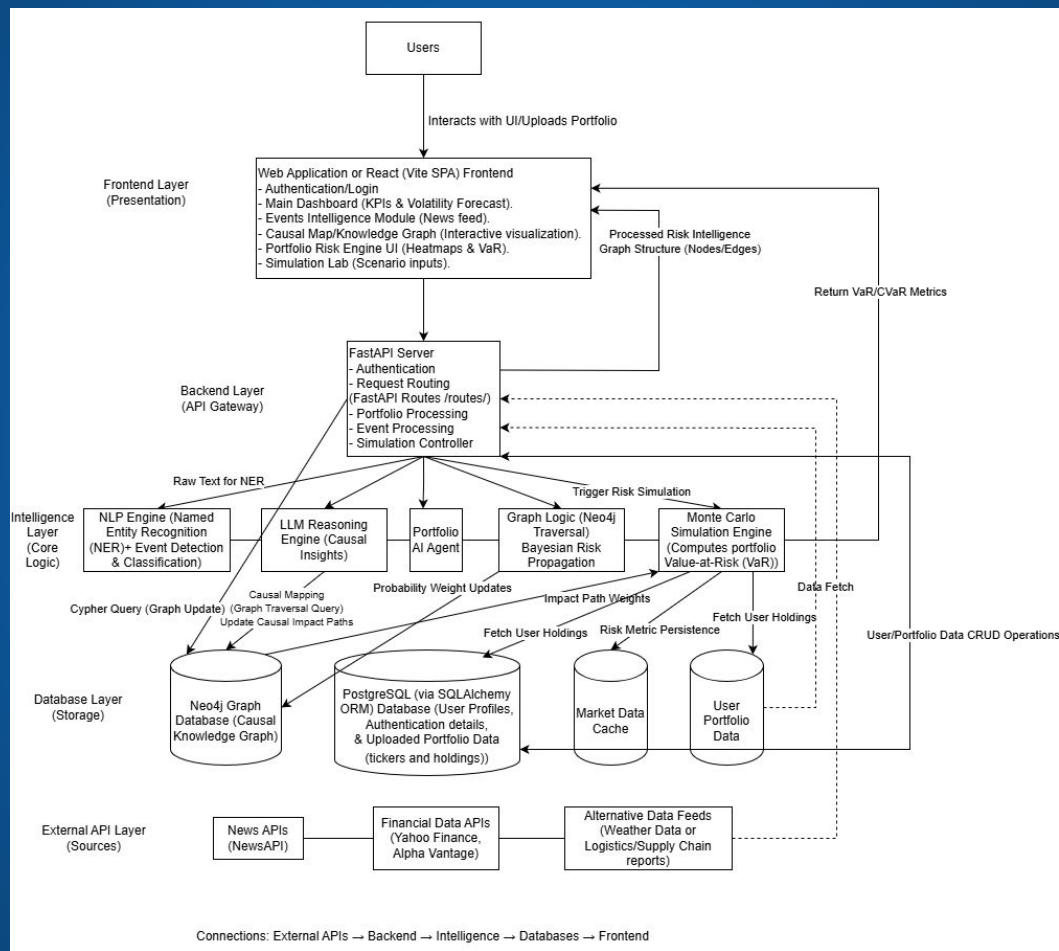
Risk Metric

Value-at-Risk is recalculated after simulation so users can compare baseline portfolio risk with event-adjusted risk

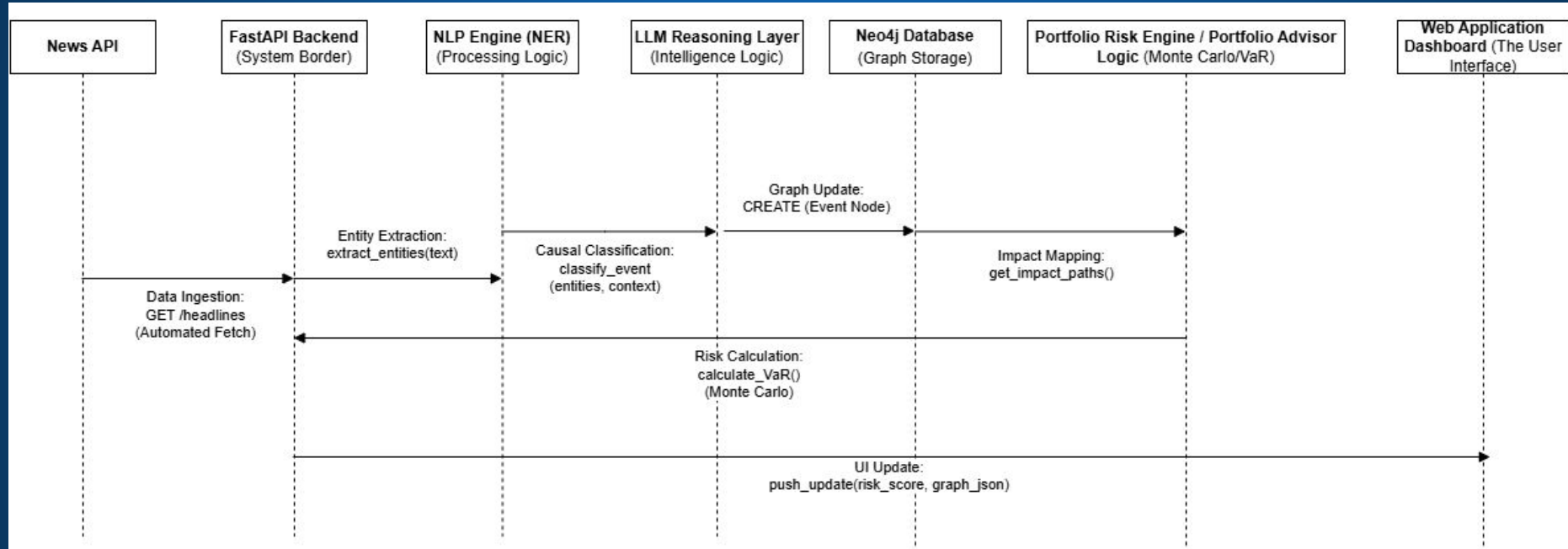
Investor Insight

The final response explains the impact in simple language, for example how a port strike can raise risk in auto-related holdings.

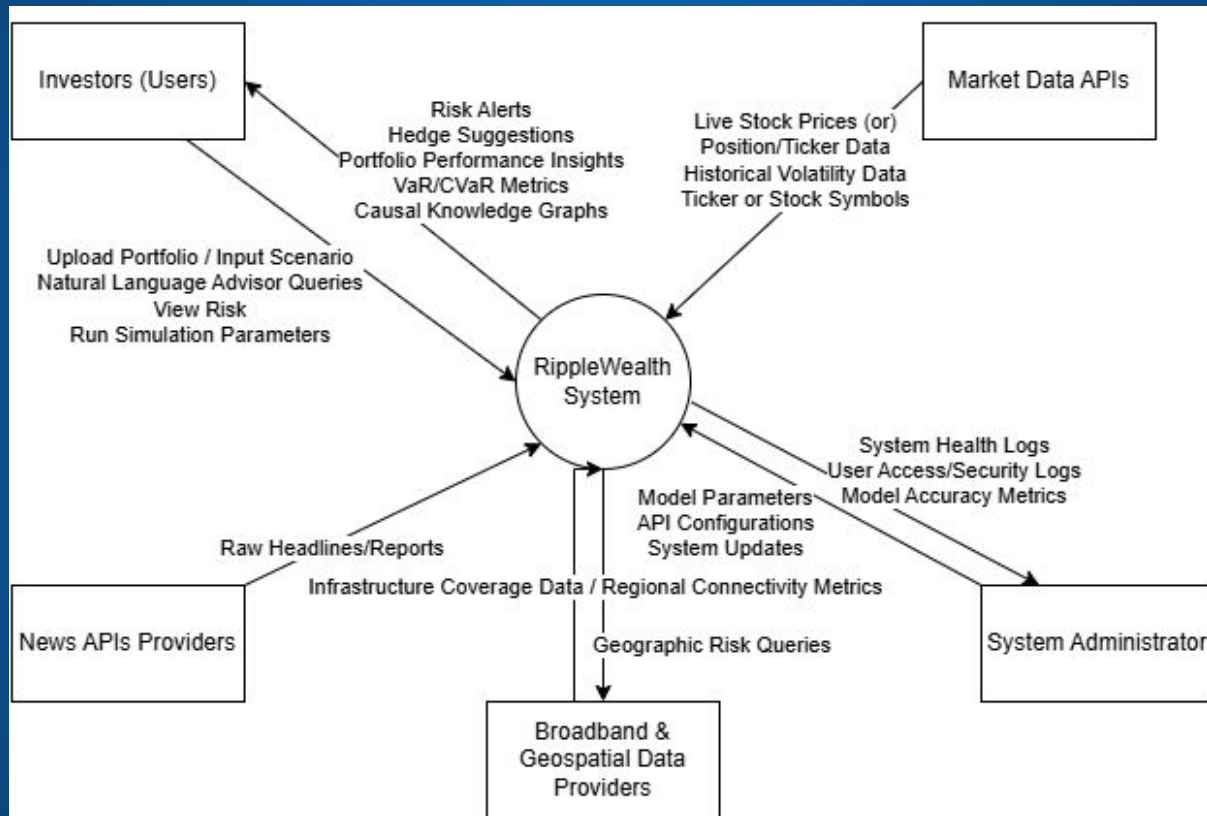
System Architecture Diagram



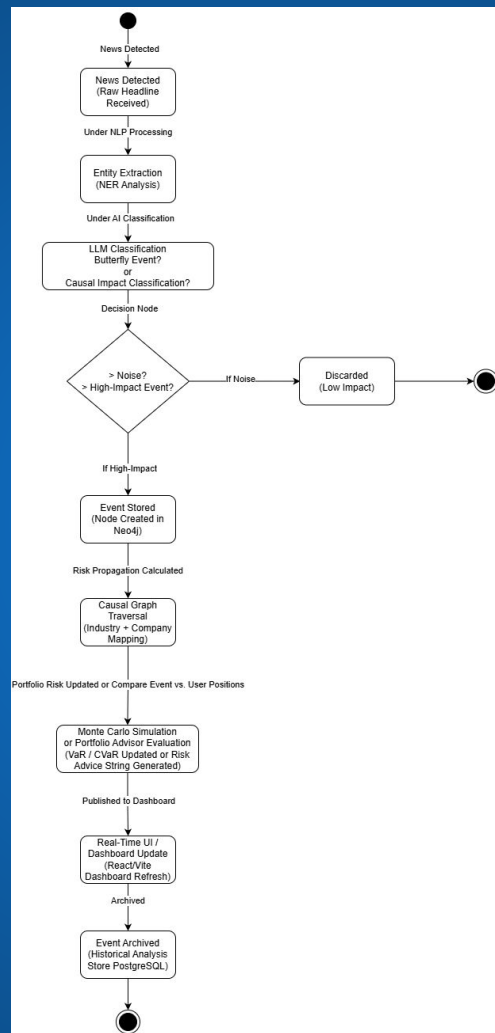
Sequence Diagram



Context Diagram



State Diagram



Sprint 1 Recap

Sprint Goal:

Build a working prototype of RippleWealth that demonstrates how global events impact portfolio risk and investment decisions.

Work Completed:

- Built core API's.
- Implemented event-driven logic for portfolio analysis.
- Connected frontend to backend using API's.

Key Features:

- Portfolio Visualization
- Global event tracking
- Risk Metrics
- Simulation engine for event impact
- Rebalancing recommendations
- Knowledge Graph

Product Backlog – User Stories

Team-8 Sprint 2

11 Mar – 7 Apr (25 work items)

65

0

0

Complete sprint

...

T8-48	As an investor, I want to see my portfolio's Value-at-Risk (VaR), so that I understand the potential losses I could face during market disruptions.	TO DO	8	
T8-54	As an investor, I want the system to calculate potential portfolio losses using Monte Carlo simulations, so that I can understand the range of possible fin...	TO DO	8	
T8-55	As an investor, I want the platform to generate explanations for its risk predictions, so that I can understand the reasoning behind the AI insights.	TO DO	8	
T8-56	As an investor, I want to upload my portfolio holdings, so that the system can analyze how global events may affect my investments.	TO DO	3	
T8-58	As an investor, I want to simulate hypothetical events like natural disasters or geopolitical crises, so that I can understand how such events might affect ...	TO DO	8	
T8-59	As a developer, I want to implement Portfolio & Risk API Endpoints, so that the frontend can securely request and receive risk intelligence data.	TO DO	5	
T8-60	As an investor, I want the platform to generate natural language explanations for its risk predictions, so that I can understand the underlying reasoning b...	TO DO	8	
T8-61	As a user, I want to view a centralized risk dashboard with a sector exposure heatmap, so that I can visually identify if my investments are too concentra...	TO DO	5	
T8-45	As a portfolio manager, I want to filter events by region, severity or industry so that I can focus only on disruptions relevant to my investments	TO DO	3	
T8-43	As as registered user, I want to recover my password, so that I can regain access to my account without losing data.	TO DO	2	
T8-42	As a returning user, I want to click Login from the landing page, so that I can securely access my dashboard.	TO DO	2	
T8-67	As a user, I want the frontend to fetch live data from backend API's, so that all the pages display dynamic and updated information	TO DO	5	
T8-68	As a user, I want quick access buttons on the home page, so that I can move directly to the feature I need.	TO DO	-	
T8-69	As a user, I want loading and error messages to appear when backend data is being fetched or fails, so that I understand the current system state.	TO DO	-	

Acceptance Criteria

Issue Type	ID	Summary	Scenario
Acceptance Criteria	T8-48	As an investor, I want to see my portfolio's Value-at-Risk (VaR), so that I understand the potential losses I could face during market disruptions.	Given my portfolio data is available When the VaR calculation is completed Then the system shows the estimated potential loss value clearly And the result is linked to my current portfolio holdings
Acceptance Criteria	T8-54	As an investor, I want the system to calculate potential portfolio losses using Monte Carlo simulations, so that I can understand the range of possible financial outcomes.	Given I am viewing my portfolio risk analysis When I request a Monte Carlo simulation Then the system runs the simulation using my portfolio data Given the simulation is completed When the results are returned Then the system displays the range of possible portfolio outcomes And the system presents estimated gains or losses across scenarios
Acceptance Criteria	T8-55	As an investor, I want the platform to generate explanations for its risk predictions, so that I can understand the reasoning behind the AI insights.	Given the system has generated a risk prediction When I view the prediction details Then the system displays an explanation of the factors contributing to that prediction Given multiple factors influence the prediction When the explanation is shown Then the system identifies the key drivers such as events, industries, or portfolio exposure
Acceptance Criteria	T8-56	As an investor, I want to upload my portfolio holdings, so that the system can analyze how global events may affect my investments.	Given the upload contains valid portfolio information When the upload is processed Then the system uses the portfolio data for risk analysis and event impact assessment Given the uploaded file is invalid or incomplete When the system validates the file Then the system displays an appropriate error message And the invalid data is not processed
Acceptance Criteria	T8-58	As an investor, I want to simulate hypothetical events like natural disasters or geopolitical crises, so that I can understand how such events might affect my investments.	Given a hypothetical event has been submitted When the simulation is executed Then the system displays the projected impact on my portfolio And the result includes estimated changes in risk or value
Acceptance Criteria	T8-59	As a developer, I want to implement Portfolio & Risk API Endpoints, so that the frontend can securely request and receive risk intelligence data.	Given the frontend sends a valid request to the Portfolio or Risk API endpoint When the request reaches the backend Then the API returns the requested portfolio or risk data in the expected format
Acceptance Criteria	T8-60	As an investor, I want the platform to generate natural language explanations for its risk predictions, so that I can understand the underlying reasoning behind the AI's insights.	Given the prediction is based on multiple data points When the explanation is shown Then the system summarizes the most important contributing factors in user-friendly language

Acceptance Criteria

Issue Type	ID	Summary	Scenario
Acceptance Criteria	T8-61	As a user, I want to view a centralized risk dashboard with a sector exposure heatmap, so that I can visually identify if my investments are too concentrated in high-risk industries.	Given I am a logged-in user When I open the dashboard Then the system displays a centralized risk dashboard Given my portfolio contains sector allocations When the dashboard loads Then the system displays a sector exposure heatmap
Acceptance Criteria	T8-45	As a portfolio manager, I want to filter events by region, severity or industry so that I can focus only on disruptions relevant to my investments	Given I am viewing the events page When I select a filter by region, severity, or industry Then the system updates the event list based on the selected filters
Acceptance Criteria	T8-43	As a registered user, I want to recover my password, so that I can regain access to my account without losing data.	Given I am a registered user on the login page When I click the Forgot Password option Then the system prompts me to enter my registered email address Given I submit a valid registered email When the password recovery request is processed Then the system sends a password reset link or instructions to my email Given I reset my password successfully When I return to the login page Then I can log in with the new password And my account data remains unchanged
Acceptance Criteria	T8-42	As a returning user, I want to click Login from the landing page, so that I can securely access my dashboard.	Given I enter valid credentials When I submit the login form Then the system authenticates me successfully And redirects me to my dashboard
Acceptance Criteria	T8-67	As a user, I want the frontend to fetch live data from backend API's, so that all the pages display dynamic and updated information	Given the API responds successfully When the data is received Then the page displays dynamic and updated information from the backend
Acceptance Criteria	T8-68	As a user, I want quick access buttons on the home page, so that I can move directly to the feature I need.	Given I click a quick access button When the action is triggered Then the system navigates me directly to the corresponding feature page
Acceptance Criteria	T8-69	As a user, I want loading and error messages to appear when backend data is being fetched or fails, so that I understand the current system state.	Given the API request fails When the system receives an error or no response Then the system displays a clear error message And the page does not crash

Test Cases

Test Case ID	Module	Test Objective	Test Steps	Expected Result
TC01	Homepage	Verify quick access buttons navigate to correct pages	1. Launch the application 2. Navigate to Home page 3. Click each quick access button one by one	User is redirected to the correct feature page for each button clicked
TC02	Simulation	Verify Monte Carlo simulation runs successfully	1. Navigate to Simulation page 2. Select portfolio 3. Start Monte Carlo simulation	System runs simulation successfully using portfolio data
TC03	Simulation	Verify Monte Carlo simulation results are displayed	1. Run Monte Carlo simulation 2. Wait for completion	System displays a range of possible portfolio outcomes with estimated gains/losses
TC04	AI Insights	Verify risk prediction explanation is generated	1. Navigate to risk prediction/insights section 2. Select a generated risk prediction	System displays explanation for the risk prediction
TC05	Events	Verify events page loads disruption events from backend	1. Navigate to Events screen	Events page loads and displays disruption events fetched from backend
TC06	Events	Verify filter by region works correctly	1. Navigate to Events page 2. Apply region filter	Only events matching selected region are displayed
TC07	API Integration	Verify Risk API endpoint returns valid response	1. Send valid request to Risk endpoint 2. Review API response	API returns risk intelligence data in expected format and status code
TC08	Loading/Error Handling	Verify error message is shown when backend request fails	1. Disconnect network or stop backend service 2. Open a page requiring API data	Clear error message is displayed and page does not crash
TC09	Events	Verify filter by industry works correctly	1. Navigate to Events page 2. Apply industry filter	Only events matching selected industry are displayed
TC10	Events	Verify filter by severity works correctly	1. Navigate to Events page 2. Apply severity filter	Only events matching selected severity are displayed



Sprint Backlog: Sprint 3

Team-8 Sprint 3 8 Apr – 5 May (7 work items)			0	0	0	Start sprint	...
T8-31	Finish Development of Frontend	TO DO ▾	-	DS			
T8-32	Finish Development of Backend	TO DO ▾	-	AM			
T8-33	Frontend-Backend Integration Working	TO DO ▾	-	SK			
T8-34	System Testing	TO DO ▾	-	JP			
T8-35	Finish Project Demo	TO DO ▾	-	JP			
T8-36	Finish Technical Paper	TO DO ▾	-	SK			
T8-37	Work on Deliverable 4	TO DO ▾	-	DS			

Stories Completed – Sprint 2

☐ Team-8 Sprint 2 11 Mar – 7 Apr (25 work items)	05134	Complete sprint
 T8-48 As an investor, I want to see my portfolio's Value-at-Risk (VaR), so that I understand the potential losses I could f...	DONE ✓	8 
 T8-54 As an investor, I want the system to calculate potential portfolio losses using Monte Carlo simulations, so that I ca...	DONE ✓	8 
 T8-55 As an investor, I want the platform to generate explanations for its risk predictions, so that I can understand the r...	DONE ✓	8 
 T8-56 As an investor, I want to upload my portfolio holdings, so that the system can analyze how global events may affe...	DONE ✓	3 
 T8-58 As an investor, I want to simulate hypothetical events like natural disasters or geopolitical crises, so that I can und...	DONE ✓	8 
 T8-59 As a developer, I want to implement Portfolio & Risk API Endpoints, so that the frontend can securely request and ...	DONE ✓	5 
 T8-60 As an investor, I want the platform to generate natural language explanations for its risk predictions, so that I can ...	DONE ✓	8 
 T8-61 As a user, I want to view a centralized risk dashboard with a sector exposure heatmap, so that I can visually ident...	DONE ✓	5 
 T8-43 As as registered user, I want to recover my password, so that I can regain access to my account without losing d...	DONE ✓	2 
 T8-42 As a returning user, I want to click Login from the landing page, so that I can securely access my dashboard.	DONE ✓	2 
 T8-67 As a user, I want the frontend to fetch live data from backend API's, so that all the pages display dynamic and up...	DONE ✓	5 
 T8-68 As a user, I want quick access buttons on the home page, so that I can move directly to the feature I need.	DONE ✓	2 

Stories Not Completed – Sprint 2

☐ ▼ Team-8 Sprint 2 11 Mar – 7 Apr (25 work items)	05134	Complete sprint
 T8-45 As a portfolio manager, I want to filter events by region, severity or industry so that I can focus only on disruption...	IN PROGRESS ▼	3 
 T8-69 As a user, I want loading and error messages to appear when backend data is being fetched or fails, so that I und...	IN PROGRESS ▼	2 

Sprint 2 Schedule

Team-8 Sprint 2

11 Mar – 7 Apr (25 work items)

0

5

134

Complete sprint

T8-20 Setup FastAPI Project & Database Models DONE 3 SK
T8-21 Implement Portfolio & Risk API Endpoints DONE 5 JP
T8-22 Causal Event Modeling DONE 8 JP
T8-23 Knowledge Graph Visualization Structure DONE 8 DS
T8-24 LLM Reasoning Logic DONE 8 AM
T8-25 Event Impact Simulation DONE 12 AM
T8-26 AI-Backend Integration DONE 5 JP
T8-27 Draft Technical Paper DONE 5 SK
T8-28 Develop Frontend DONE 8 DS
T8-29 Develop Backend DONE 5 AM
T8-30 Work on Deliverable 3 DONE 3 SK

Metrics - Status Overview as of Sprint 2

Spaces

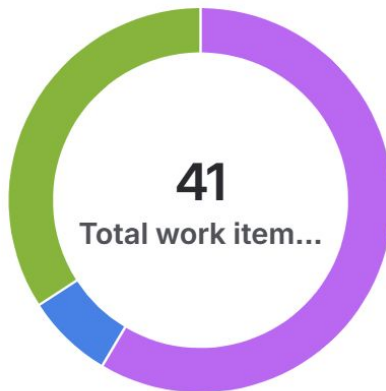
 **RippleWealth**  ...



 Summary  Timeline  Backlog  Board  Calendar  List  Forms  Development  Code More 4 +

Status overview

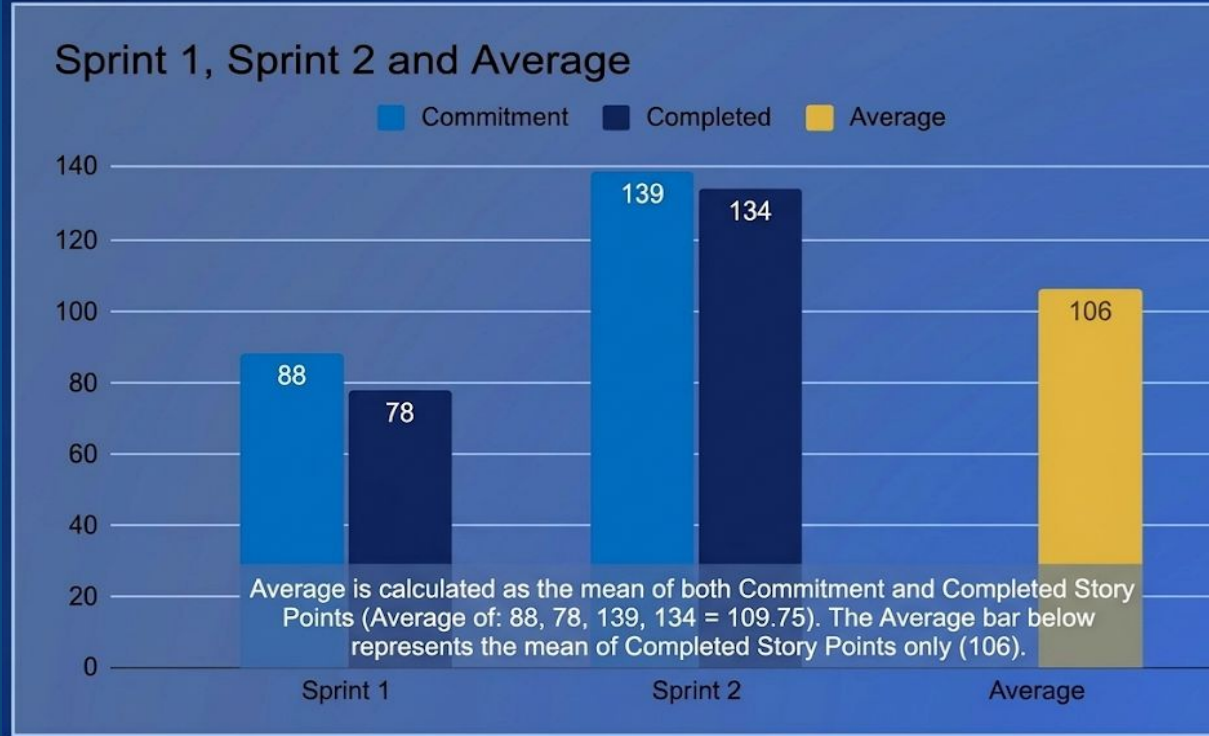
Get a snapshot of the status of your work items. [View all work items](#)



-  Done: 24
-  In Progress: 3
-  To Do: 14

Metrics - Team's Historical Velocity (Average)

Team History Velocity (Average)



Metrics – Sprint 2 Burndown Chart

Spaces / RippleWealth / Reports

Sprint burndown chart

> [How to read this report](#)

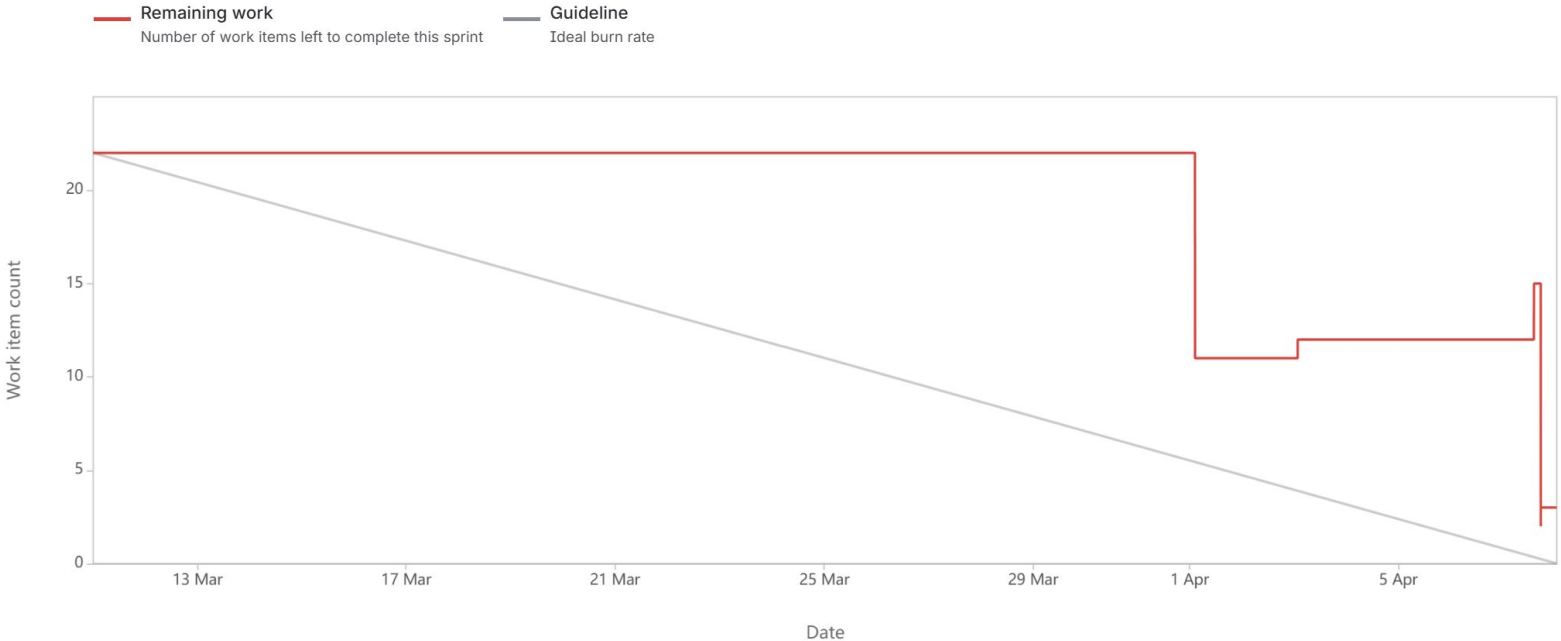
Sprint

Team-8 Sprint 2

Estimation field

Work item count

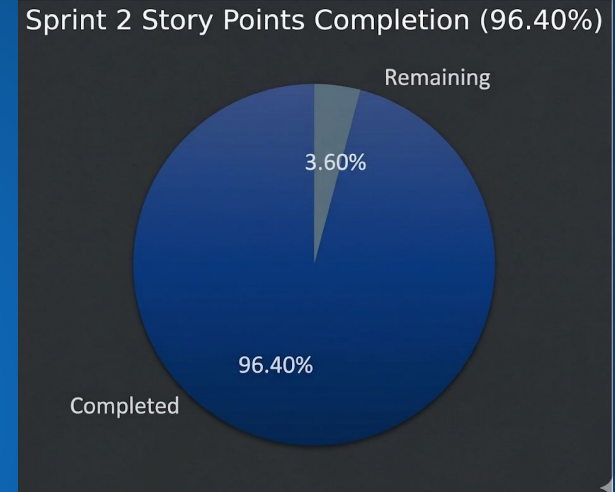
Date - 11 March 2026 - 7 April 2026



Metrics – Sprint 2 Ratio Status

Committed Stories Points: 139

Completed Stories Points: 134



Completed/Committed Ratio (Story Points) = $134/139$, which is 96.40%

Retrospective



What Went Well

- ❖ **Architectural Pivot:** Successfully transitioned the frontend from Streamlit to React (Vite) to support high-performance Knowledge Graph visualizations.
- ❖ **Core Backend & Data:** Established a robust FastAPI structure with functional routes for portfolio management and risk analytics.
- ❖ **Causal Reasoning:** Integrated Neo4j for graph-based impact tracing, allowing the system to map "Butterfly Events" across global industries.
- ❖ **Data Enrichment:** Leveraged Python logic to incorporate geospatial broadband metrics, adding a unique infrastructure-risk dimension to scoring.
- ❖ **Team Synergy & Efficient Execution:** Maintained strong collaboration and effective pair programming despite a remote environment, leading to increased team bonding.

Retrospective



What Could Be Improved

- ❖ **Coding Redundancies & Sync:** Identified overlapping logic between FastAPI routes and AI agents. A centralized utility structure and unified schema are needed to eliminate redundancy and resolve Pydantic-to-React mapping delays.
- ❖ **Mandatory Meeting Alignment:** Presentation deliverables, sprint goals, and detailed “work-done” updates should be mandatory agenda items in every meeting to ensure team-wide alignment.
- ❖ **Task Granularity:** Complex technical tasks, such as LLM Reasoning and Causal Modeling, require finer sub-task breakdowns for more accurate effort estimation and tracking.
- ❖ **Communication Flow:** Standardizing status update frequency during meetings will help identify and resolve technical blockers early, preventing delays in the final demo.

Retrospective



Action Items for Next Sprint

- ❖ **Process Rigor:** Institute mandatory weekly technical summaries and mid-sprint "blocker" check-ins to ensure deliverable alignment.
- ❖ **Planning & Availability:** Establish better meeting windows and ensure all stories have clear Acceptance Criteria before development begins.
- ❖ **Velocity Adjustment:** Plan for fewer story points to improve team velocity and allow for iterative presentation updates.
- ❖ **Performance & Docs:** Allocate dedicated time for modular code refactoring and optimization alongside "Documentation Power Hours" to finalize the Technical Paper and GitHub Wiki.

Sprint 3 Schedule

☐ **Team-8 Sprint 3** 8 Apr – 5 May (7 work items)

0 0 0 Start sprint ...

 T8-31	Finish Development of Frontend	TO DO ▾	-	
 T8-32	Finish Development of Backend	TO DO ▾	-	
 T8-33	Frontend-Backend Integration Working	TO DO ▾	-	
 T8-34	System Testing	TO DO ▾	-	
 T8-35	Finish Project Demo	TO DO ▾	-	
 T8-36	Finish Technical Paper	TO DO ▾	-	
 T8-37	Work on Deliverable 4	TO DO ▾	-	

Stories planned and committed for Sprint 3

Team-8 Sprint 3

8 Apr – 5 May (14 work items)

6400Start sprint...

T8-70

As an investor, I want to complete the full workflow from portfolio upload to risk insights, so that I can evaluate th...

TO DO

8

T8-71

As a user, I want secure session handling (login/logout/session timeout), so that my account remains protected.

TO DO

5

T8-72

As a user, I want the application to work across different browsers, so that I can access it from any environment.

TO DO

3

T8-73

As a user, I want a sidebar or menu that works on different screen sizes so I can navigate easily.

TO DO

3

T8-74

As a user, I want to see a loading spinner while simulations run so I know the system is processing.

TO DO

2

T8-75

As an investor, I want to search for specific stocks in my uploaded portfolio so I can check their individual risk sc...

TO DO

3

T8-76

As a user, I want quick access buttons on the home page for direct navigation.

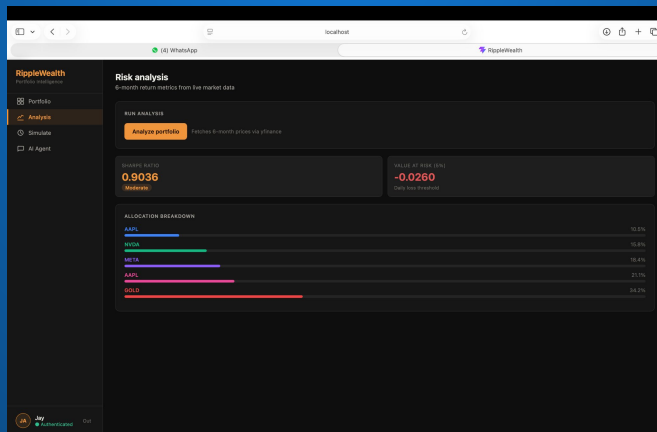
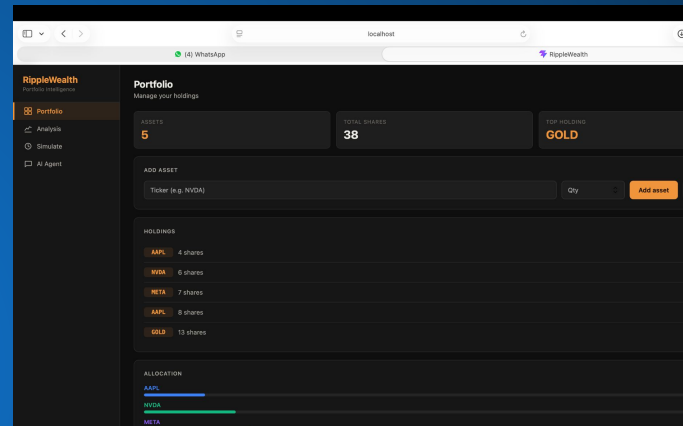
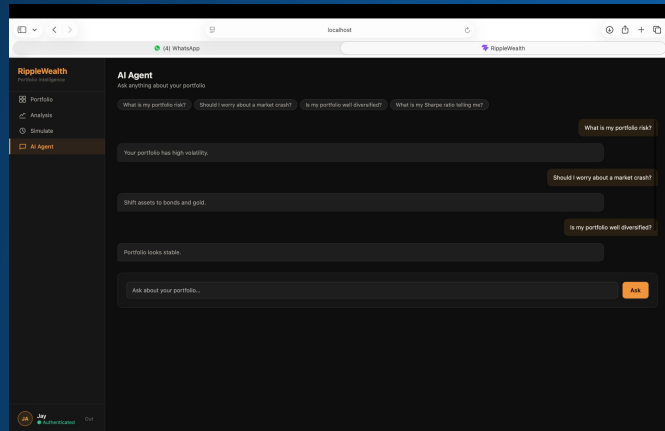
TO DO

2

Project Demo – Sprint 2 (Current Sprint)



Application Screenshots



Working API Screenshots

This screenshot shows a Swagger UI interface for an API. The top navigation bar includes tabs for 'RippleWealth', 'CS Capstone Project - Google Drive', 'Sprint 2 Presentation.pptx - Google Slides', and 'FastAPI - Swagger UI'. The main content area displays a parameter named 'user_id' of type 'integer' and 'required', with a text input field containing 'user_id'. Below the parameter, there are two response sections. The first response is a 'Successful Response' with a status code of 200, showing a media type of 'application/json' and an example value of 'string'. The second response is a 'Validation Error' with a status code of 422, showing a media type of 'application/json' and an example value of a JSON object:

```
{  "detail": [    {      "loc": [        "string",        0      ],      "msg": "string",      "type": "string",      "input": "string"    }  ]}
```

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This screenshot shows a Swagger UI interface for an API, specifically for a 'default' endpoint. The top navigation bar includes tabs for 'RippleWealth', 'CS Capstone Project - Google Drive', 'Sprint 2 Presentation.pptx - Google Slides', and 'FastAPI - Swagger UI'. The main content area displays a 'POST /register Register' endpoint. The 'Parameters' section shows two required parameters: 'username' of type 'string' and 'password' of type 'string', both with text input fields. Below the parameters, there are two response sections. The first response is a 'Successful Response' with a status code of 200, showing a media type of 'application/json' and an example value of 'string'. The second response is a 'Validation Error' with a status code of 422, showing a media type of 'application/json' and an example value of a JSON object:

```
{  "detail": [    {      "loc": [        "string",        0      ],      "msg": "string",      "type": "string",      "input": "string"    }  ]}
```

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Live Application Demo

GitHub Wiki Page Link



[SharvaniKadarla/RippleWealth Wiki Page \(github.com\)](https://github.com/SharvaniKadarla/RippleWealth/wiki)

View on GitHub

Technical Paper Link



[RIPPLEWEALTH TECHNICAL PAPER \(GITHUB.COM\)](#)

View on GitHub

THANK YOU